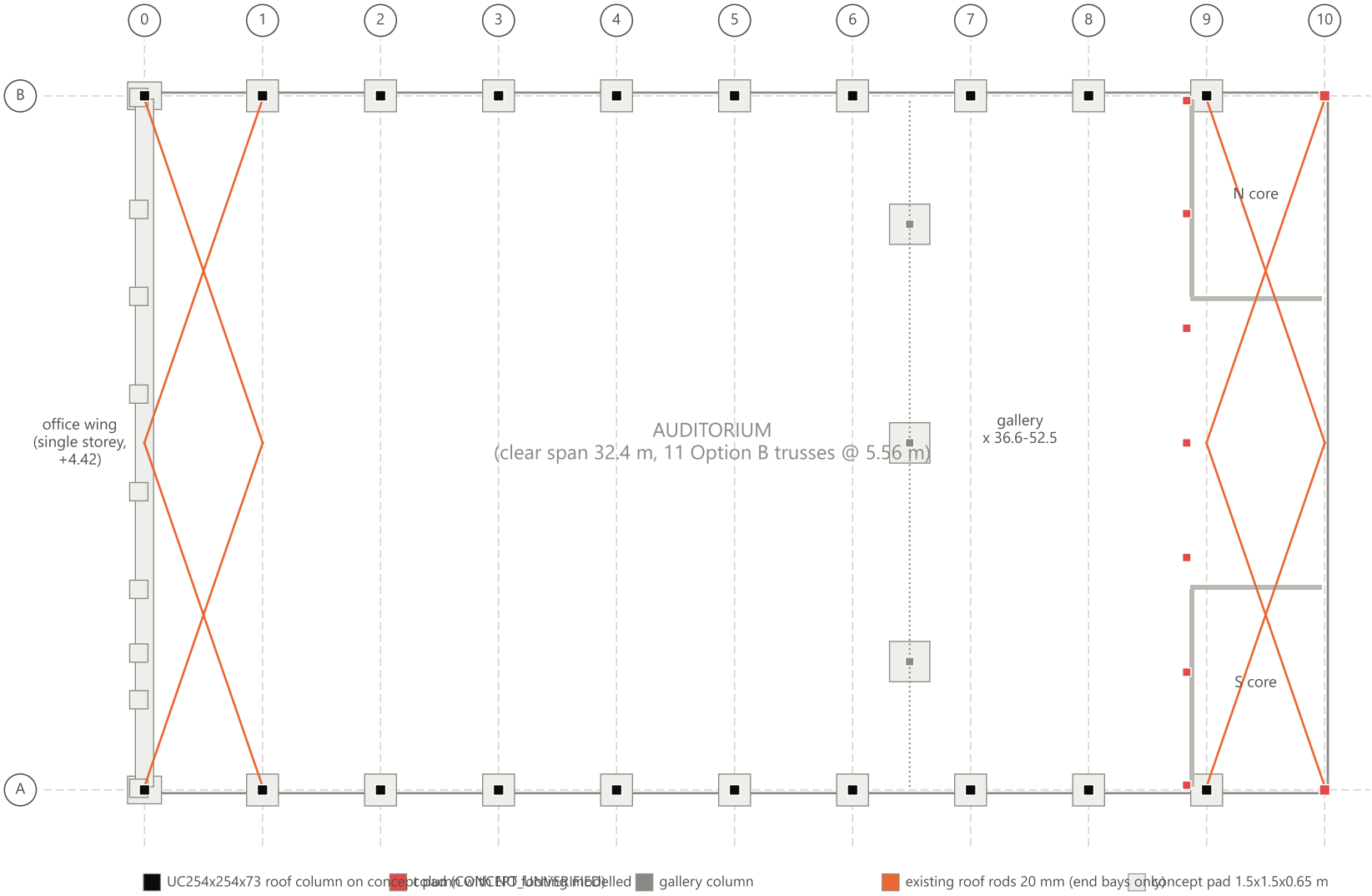
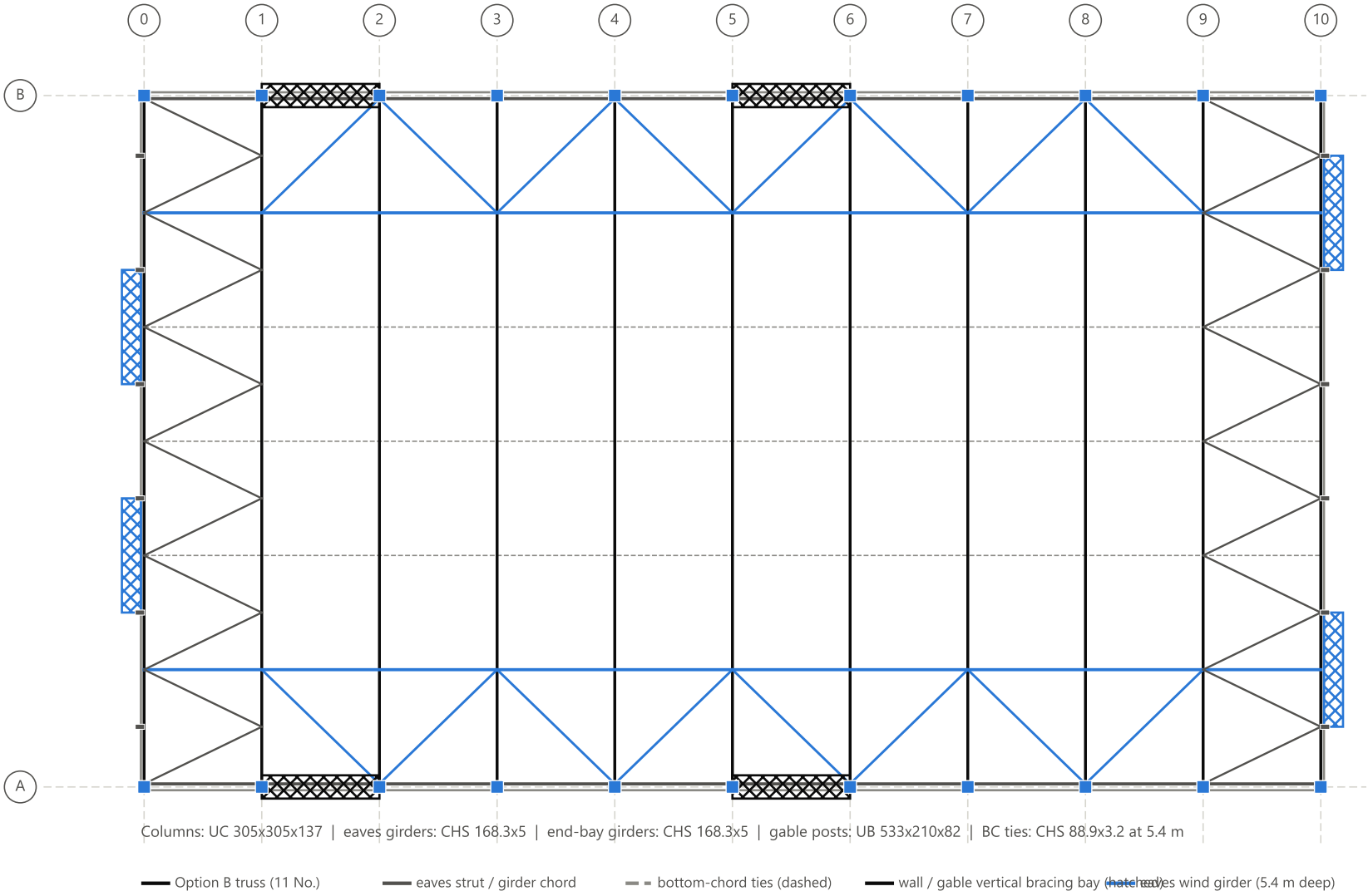
S-01 Existing structural model: roof columns, footings and bracing (IFC, read-only)



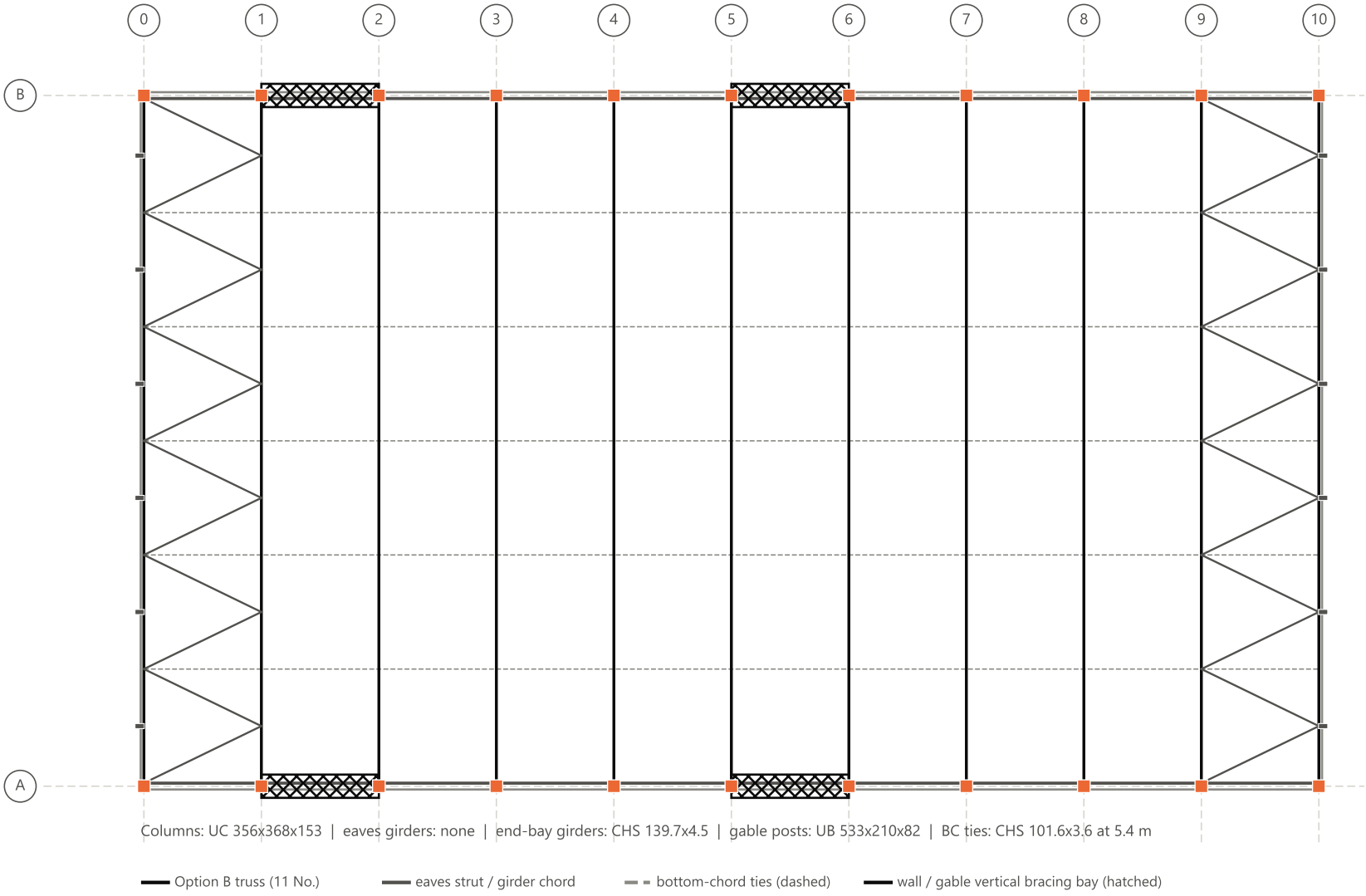
PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-02 Roof-plane framing and stability plan - S1 braced: pinned bases, eaves wind girders, gable bracing



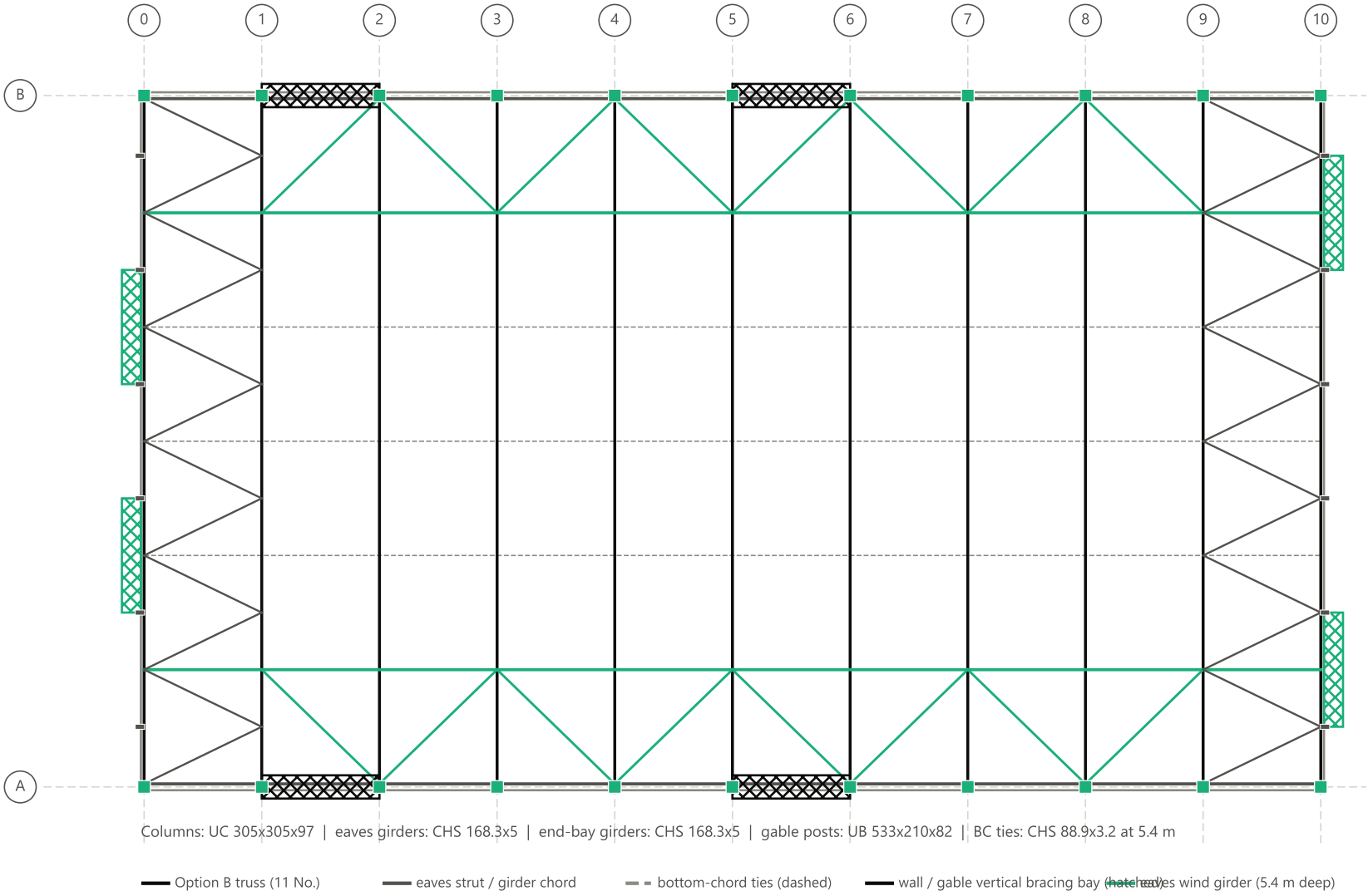
PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-03 Roof-plane framing and stability plan - S2 moment frames: fixed bases + truss heel knee at every line



PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-04 Roof-plane framing and stability plan - S3 hybrid: fixed-base propped columns, eaves wind girders, gable bracing



PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-05a Side-wall elevation: column grid, openings and longitudinal bracing (all schemes)

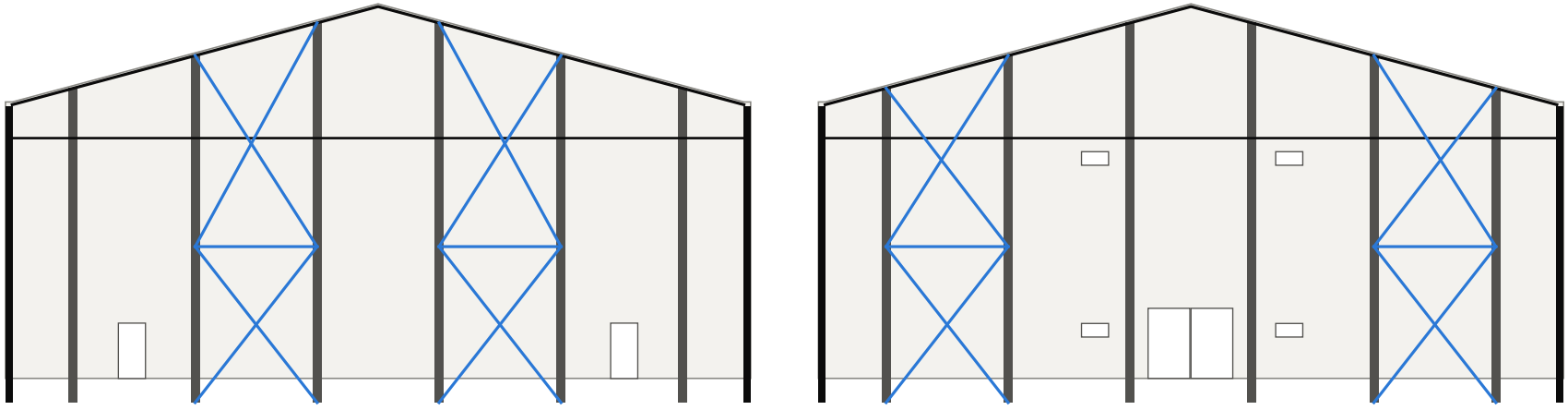
South side wall (north mirror): 300 mm masonry infill between columns, clerestory windows +10.10 to +11.80, doors at bays 0, 4, 7, 9.
Longitudinal bracing in bays 1 and 5 only: X below the clerestory sill (+9.90) with knee braces inside the 1.08 m window piers.



S-05b Gable elevations: wind posts and transverse bracing

West gable (behind stage), line 0

East gable (stair cores), line 10

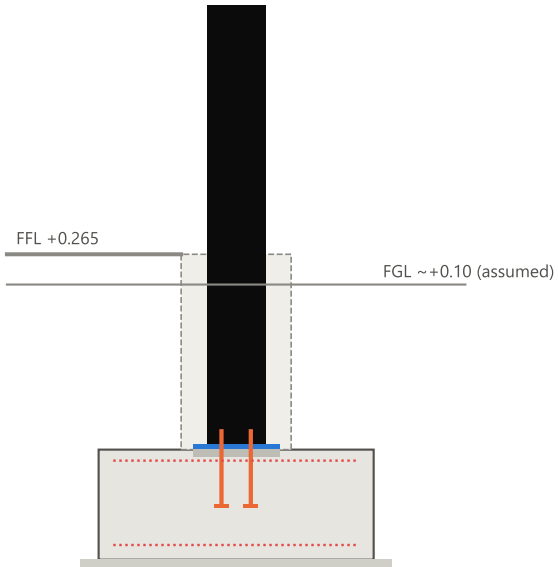


Gable wind posts (6 No. per gable at 5.4 m, clear of doors and the east glazed slot) span base to top chord. Blue: transverse gable bracing for S1/S3 (two-tier X). S2 needs posts but no gable bracing.

PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

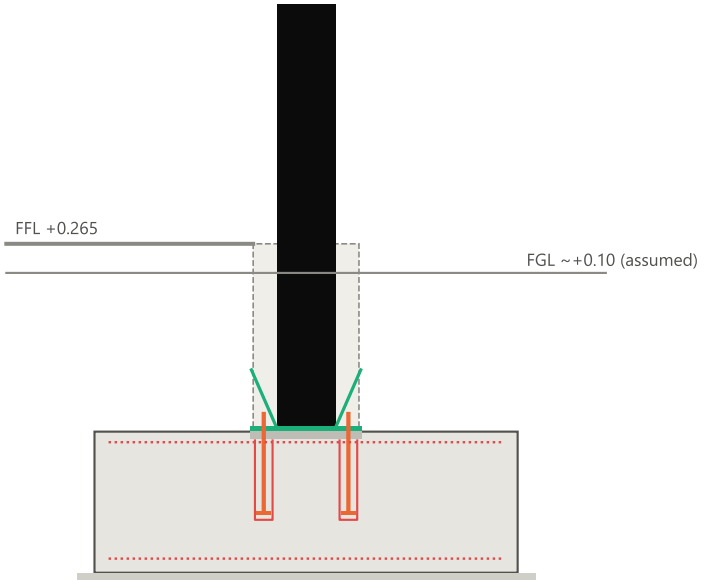
S-06 Column base and pad concept sections (indicative; anchorage and reinforcement to be designed)

Pinned base (S1)



Column UC 305x305x137
Base plate 470 x 460 x 25 mm S355 on 40-50 mm non-shrink grout
Holding-down: 4 M24 8.8 cast-in, hef 300 mm
Pad 1.5 x 1.9 x 0.60 m C25/30, G2 soil, F2 arrangement
Bottom steel ~780 mm2/m each way; top ~0 mm2/m
Column encased in C25/30 from pad to slab (corrosion protection below FFL)

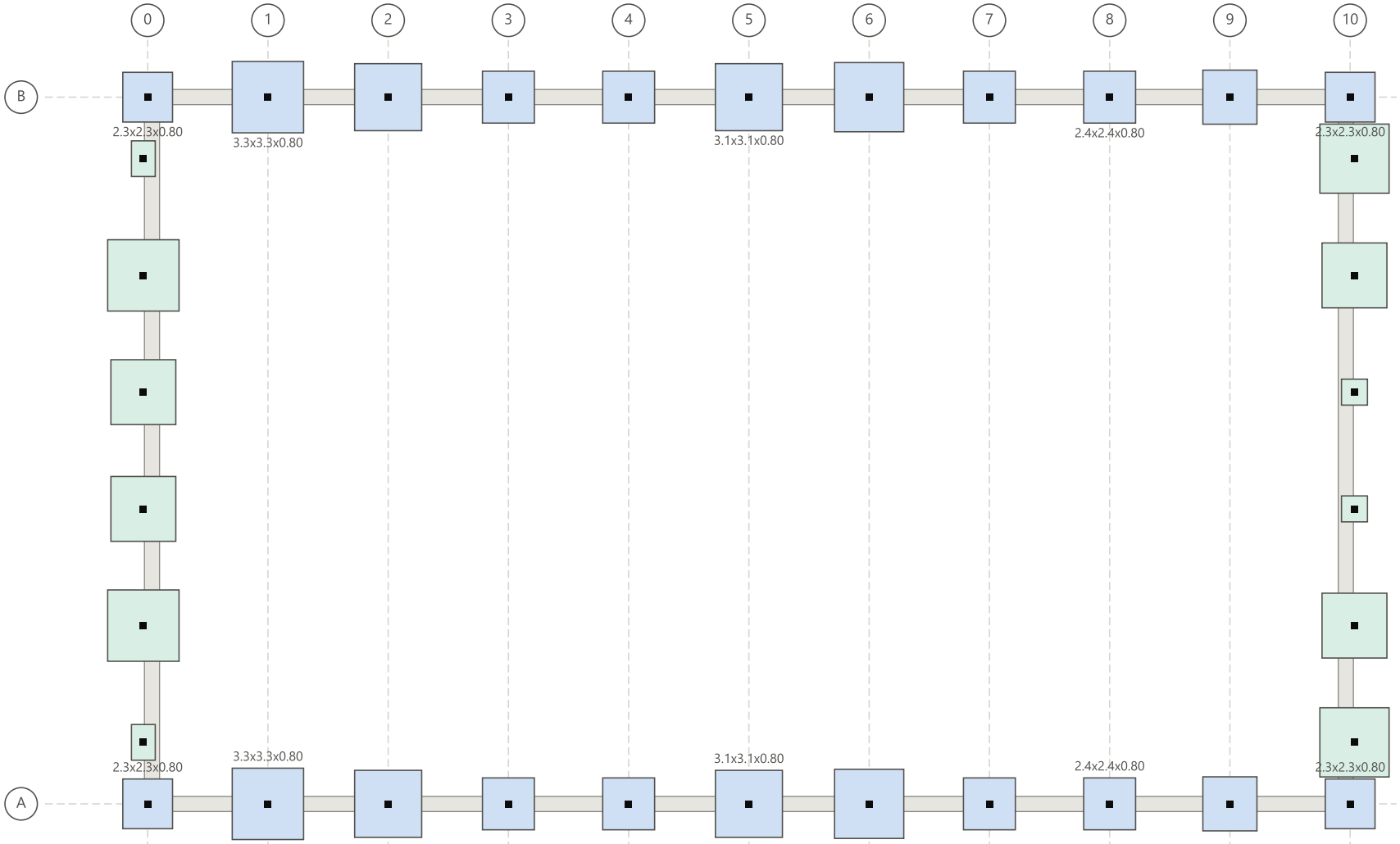
Fixed base (S3 recommended)



Column UC 305x305x97
Base plate 608 x 455 x 30 mm S355 on 40-50 mm non-shrink grout
Holding-down: 6 M24 8.8 cast-in with anchor plates, hef 450 mm
Pad 2.4 x 2.4 x 0.80 m C25/30, G2 soil, F2 arrangement
Bottom steel ~1040 mm2/m each way; top ~624 mm2/m
Column encased in C25/30 from pad to slab (corrosion protection below FFL)
Anchor (supplementary) reinforcement: hairpins around tension bolts
Stiffened base: 4 gussets (concept)

PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-07 Foundation concept plan - recommended S3 hybrid (indicative)

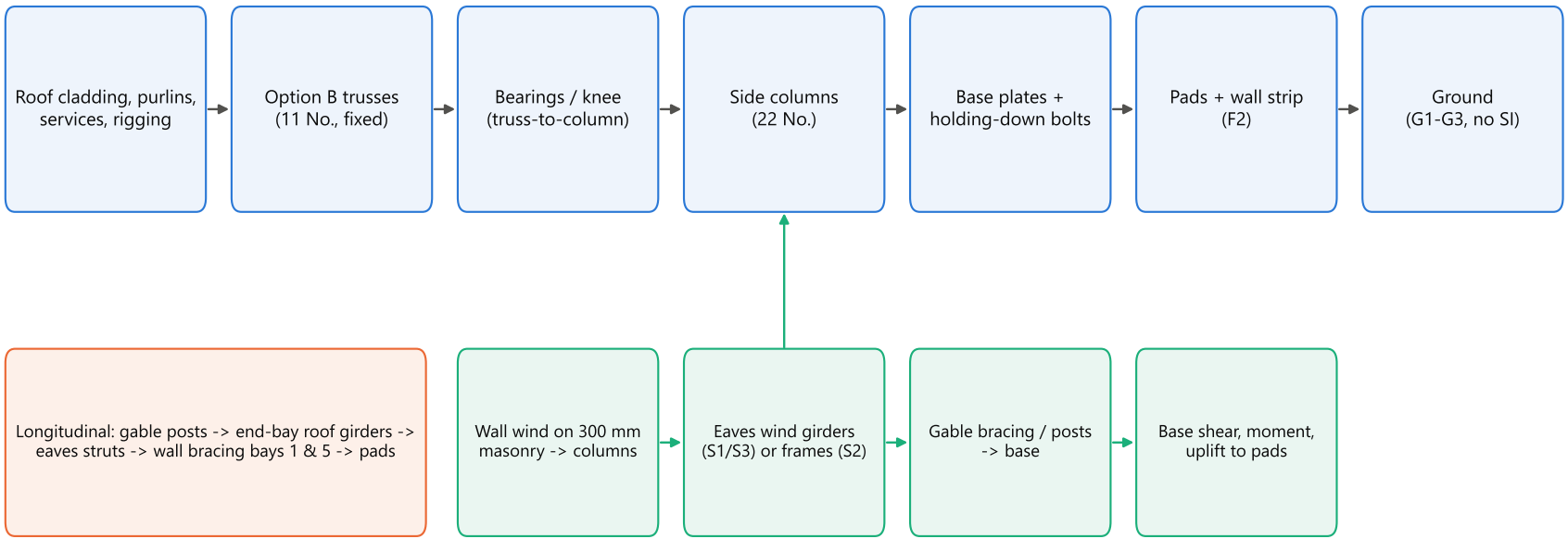


F2 arrangement, G2 soil (150 kPa assumed): column pads cast integral with a reinforced continuous strip under the masonry walls; gable post pads green. Sizes indicative only - no site investigation.

PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

S-08 Structural load path

Gravity (blue) and lateral (green / orange) load paths, Option B roof to ground



PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION